# Yibin Fan

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## Education

**North Carolina State University** — Raleigh, NC
*Master of Computer Science* — *Aug. 2025 – May. 2027*
GPA: 4.0/4.0

**Chengdu University of Technology** — Chengdu, China
*Bachelor of Engineering in Computer Science* — *Sep. 2021 – Jun. 2025*
GPA: 3.6/5.0, Top 10%

## Research Interests

Reinforcement Learning, Decision-Making under Partial Observability, Multimodal Reasoning, Embodied AI

## Selected Publications

**Probe–Exploit: When and How to Recover Hidden Symmetry Breaker in Equivariant RL**
**Yibin Fan**, Qi Zhang
*NeurReps 2026: Symmetry and Geometry in Neural Representations (NeurIPS Workshop)*

## Research Experience

**Research Intern** — Jun. 2026 – Aug. 2026
*Worcester Polytechnic Institute* — *Remote*

- Investigated how agents should balance information acquisition and action under partial observability in controlled GridWorld environments with hidden, episode-dependent transition dynamics.
- Developed an equivariant reinforcement learning framework that uses active system identification to recover task-relevant context for adaptive policy learning.
- Applied value-of-information analysis to characterize when targeted information gathering can improve decision-making under limited interaction budgets.

**Graduate Research Assistant** — Sep. 2025 – Jan. 2026
*North Carolina State University* — *Raleigh, NC*

- Developed deep learning models to predict shifts in RNA subcellular localization across biological conditions and time points using sequence and structural features.
- Built a reproducible pipeline for data processing, model training, and experimental evaluation.

**Undergraduate Research Assistant** — Jul. 2024 – Oct. 2024
*Chengdu University of Technology* — *Chengdu, China*

- Investigated deep learning approaches for medical image segmentation, focusing on data preprocessing and model development.
- Implemented and evaluated segmentation pipelines to compare model performance across experimental settings.

## Selected Research Project

**Rule-Grounded Medical Agent for Traceable Multimodal Diagnosis** — Sep. 2025 – Nov. 2025

- Developed a multimodal diagnostic agent combining visual evidence, LLM reasoning, and structured tool use.
- Designed a traceable reasoning architecture linking evidence, tool execution, intermediate findings, and diagnostic conclusions.
- Integrated confidence estimation with LLM-based prompt refinement to support stable and interpretable reasoning.